

Starting from a 100% U.S. portfolio (the square in Figure 3.2), an investor can improve her risk–return trade-off by including Japanese equities. This moves her position from the United States (the square) to Japan (the circle) and the investor moves upward along the frontier in a clockwise direction. Portfolios to the right-hand side of the circle (the 100% Japan position) represent *levered* portfolios. Portfolios on the top half of the frontier past the circle are constructed by shorting the United States, like a –30% position, and then investing the short proceeds in a levered Japanese position, which would be 130% in this case. All the efficient portfolios lying on the top half of the frontier—those portfolios with the highest returns for a given level of risk—contain Japanese equities. The minimum variance portfolio, which is the left-most tip of the mean-variance bullet, also includes Japan.

The American investor can improve her risk–return trade-off by holding some of Japan because Japan provides *diversification benefits*. This is the fundamental concept in mean-variance investing, and it corresponds to the common adage “don’t put all your eggs in one basket.” The United States and Japan held together are better than the United States held alone. Owning both protects us from the catastrophe that one individual investment will be lost. The advantages of diversification imply that we cannot consider assets in isolation; we need to think about how assets behave together. This is the most important takeaway of this chapter.

Diversified, efficient portfolios of the United States and Japan have higher returns and lower risk than the 100% U.S. position. Why? When the investor combines the United States and Japan, the portfolio reduces risk because when one asset does poorly, another might do well. It’s like buying insurance (except that the purchaser of insurance loses money, on average, whereas an investor practicing diversification makes more money, on average). When the United States does relatively poorly, like during the 1980s, Japan has a possibility of doing well. Some of the risk of the U.S. position is avoidable and can be offset by holding Japan as insurance.

What about the opposite? During the 1990s Japan was in the doldrums and the United States took off. Hindsight tells us that the U.S. investor would have been better off holding only the United States. Yes, he would—*ex post*. But forecasting is always hard. At the beginning of the 1990s, the investor would have been better off on an *ex-ante* basis by holding a portfolio of both the United States and Japan. What if the roles were reversed so that in the 1990s Japan did take over the world, and the U.S. swapped places with Japan? Holding Japan in 1990 diversified away some of this *ex-ante* risk.

The formal theory behind diversification was developed by Harry Markowitz (1952), who was awarded the Nobel Prize in 1990. The revolutionary capital asset pricing model (CAPM) is laid on the capstone of mean-variance investing, and we discuss that model in chapter 6. The CAPM pushes the diversification concept further and derives that an asset’s risk premium is related to the (lack of) diversification benefits of that asset. This turns out to be the asset’s beta.